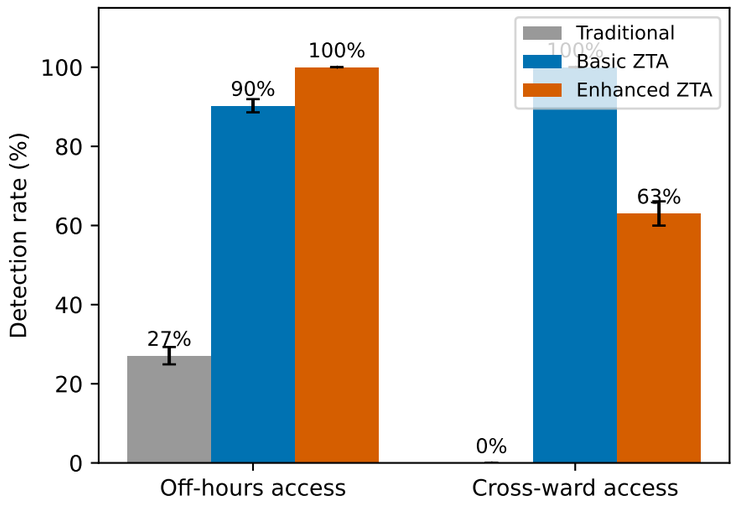
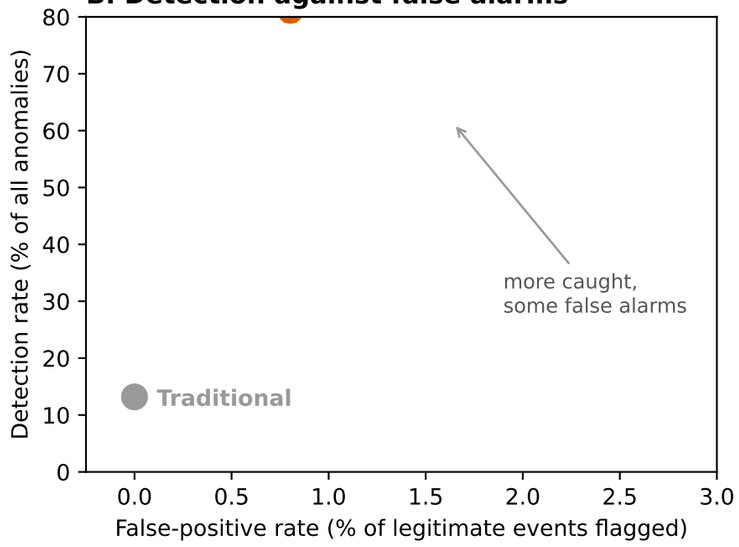
Enhanced Zero Trust vs Basic Zero Trust vs Traditional security

Results on 20 independent synthetic 30-day access logs (Isolation Forest trained once, never retrained)

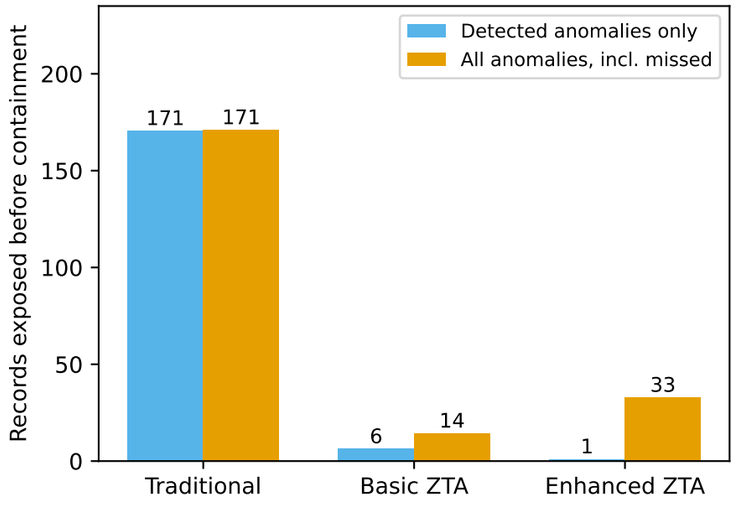
A. Which anomalies each condition catches



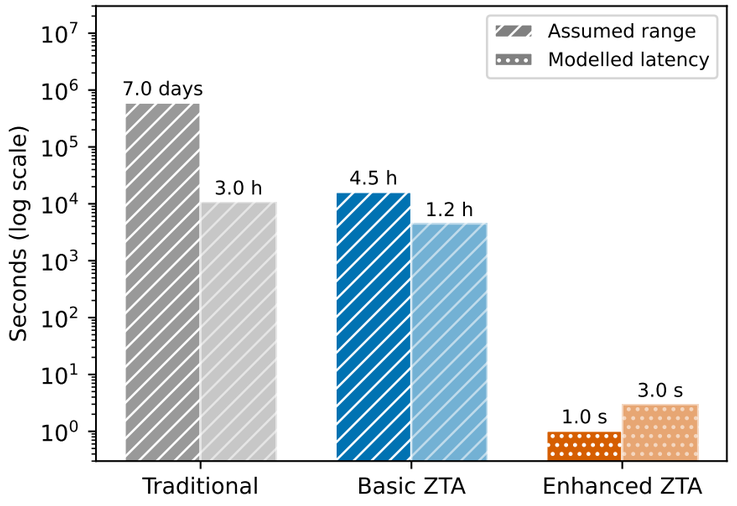
B. Detection against false alarms



C. Patient records exposed (modelled)



D. Time to detect (dark) and respond (light)



Headline metrics (mean over the datasets)

	Traditional	Basic ZTA	Enhanced ZTA
Detection: off-hours	27.1%	90.3%	100.0%
Detection: cross-ward	0.0%	100.0%	63.0%
Detection: overall	13.2%	95.2%	81.2%
False-positive rate	0.0%	1.2%	0.8%
Precision	100.0%	80.7%	84.4%
F1 score	23.3%	87.3%	82.7%
Accuracy*	95.6%	98.6%	98.3%
Time to detect (MTTD)†	7.0 days	4.5 h	1.0 s
Time to respond (MTTR)†	3.0 h	1.2 h	3.0 s
Records exposed (detected)†	171	6	1
Records exposed (incl. missed)†	171	14	33

Green = best value in the row. Overall detection pools both scenarios.
* Accuracy is high for every condition: a system that flags nothing scores 94.9%.
† Traditional and Basic ZTA timings are assumed ranges; Enhanced ZTA's are modelled latencies. Not measurements.

Synthetic data (Synthea patients, simulated access log with injected off-hours and cross-ward anomalies). Results show what the design achieves under the stated assumptions, not performance in a real hospital.